

Haiqin Wang

Department of Chemistry, Harvey Mudd College, 301 Platt Blvd., Claremont, CA 91711, USA

Email: cloverwhq@gmail.com

Phone: 086-13810329150

ORCID: orcid.org/0000-0002-4969-220X

Google Scholar: [Haiqin Wang](#)

RESEARCH INTERESTS

- Theory and simulation of polymer solutions
- Biological and soft matter Physics
- Statistical mechanics and non-equilibrium fluctuations

EDUCATION

Technion - Israel Institute of Technology, Haifa, Israel

Ph.D., Department of Physics

Dissertation Title: “”

Oct. 2018 — May. 2024

Advisors: Prof. Xinpeng Xu and Prof. Yariv Kafri

Hong Kong University of Science and Technology, Hongkong, China

Visiting Student, Department of Mathematics

Oct. 2016 — Mar. 2018

Advisor: Prof. Tiezheng Qian

Beijing Normal University, Beijing, China

M.Sc., Department of Physics

Dissertation Title:

Sep. 2015 — Jul. 2018

Advisor: Prof. Dadong Yan

Capital Normal University, Beijing, China

B.Sc., Department of Physics

Dissertation Title: “The Edge Effect of Finite Honeycomb Lattice Ising Model”

Sep. 2011 — Jul. 2015

Advisor: Prof. Yunsong Zhou

APPOINTMENTS

Harvey Mudd College, Claremont, USA

Postdoc, Department of Chemistry

Feb. 2025 — Present

Advisor: Prof. Bilin Zhuang

Wenzhou Institute, Zhejiang, China

Research Assistant

Mar. 2024 — Dec. 2024

Advisors: Prof. Masao Doi & Prof. Shigeyuki Komura

TEACHING

Physic 2 Tutorial at Guangdong Technion Israel Institute of Technology

2019, 2020 (Virtual), 2023

Chem 161: Classical and Statistical Thermodynamics Co-teach at Harvey Mudd College

2026 Spring

PUBLICATIONS

[†] Authors contribute equally, *Corresponding authors

- **H. Wang**, T. Qian, X. Xu*, Onsager’s variational principle in active soft matter, *Soft Matter*, **17**, 3634-3653 (2021)
- **H. Wang**, X. Xu*, Continuum elastic models for force transmission in biopolymer gels, *Soft Matter*, **16**, 10781-10808 (2020)
- **H. Wang**, B. Zou, J. Su, D. Wang, X. Xu* Variational methods and deep Ritz method for active elastic solids, *Soft Matter*, **18**, 6015-6031 (2022)
- **H. Wang**, X. Xu*, Variational approximation methods for long-range force transmission in biopolymer gels, *Chin. Phys. B*, **31** (10), 104602 (2022)

- X., Yu[†] (Exp.), **H. Wang[†]** (Theory), F. Ye, X. Wang*, Q. Fan*, X. Xu*, Biphasic curvature-dependence of cell migration inside microcylinders: persistent randomness versus directionality, *Biophys. J.* **124**(23), 4176-4188 (2025).
- **H. Wang**, D. Yan, T. Qian*, A phenomenological approach to the deposition pattern of evaporating droplets with contact line pinning, *Journal of Physics: Condensed Matter*, **30**, 435001 (2018)
- J. Su[†], **H. Wang[†]**, Z. Yan, X. Xu*, Active laminated-plate model for spontaneous bending of Hydra tissue fragments driven by supracellular actomyosin bundles, *Commun. Phys.* **6**, 185. (2023)
- Y. Wen[†], Z. Li[†], **H. Wang**, J. Zheng, J. Tang, P. Lai*, X. Xu*, P. Tong*, Activity-assisted barrier-crossing of self-propelled colloids over parallel microgrooves, *Phys. Rev. E*, **107**, L032601 (2023)
- Z. Li[†], B. Zou[†], **H. Wang**, J. Su, D. Wang*, and X. Xu*, Deep learning-based computational method for soft matter dynamics: Deep Onsager-Machlup method [arXiv:2308.14513](https://arxiv.org/abs/2308.14513), *Communications in Computational Physics*, accepted
- S. Bose, **H. Wang**, X. Xu*, A. Gopinath* and K. Dasbiswas*, Elastic interactions compete with persistent cell motility to drive durotaxis, *Biophys. J.*, **123**(21), 3721-3735(2024)
- **H. Wang[†]**, Z. Shen[†], X. Xu*, Phenomenological Modeling of Persistent Random Cell Migration: A Mini-review, in preparation

PRESENTATIONS

- **Berkeley Statistical Mechanics Meeting, University of California, Berkeley, CA, USA** Jan. 8-11, 2026
Hybrid Adaptive Particle-Continuum Simulation Method for Aqueous Solution [Poster]
- **Frontiers in Soft Matter and Macromolecular Networks, University of San Diego, CA, USA** Sep. 9, 2025
Flow-driven translocation of macromolecules across the slit diaphragm in the kidney glomerulus [Poster]
- **APS Global Physics Summit, Anaheim, CA, USA** March 16-21, 2025
Persistent random walk: a phenomenological paradigm for cell migration on solid substrates [Poster]
- **Boulder Summer School 2024: Self-Organizing Matter, Boulder, USA** July 1-26, 2024
Persistent random walk: a phenomenological paradigm for cell migration on solid substrates [Poster]
- **Persistent random walk: a phenomenological paradigm for cell migration on solid substrates** Jan. 2024
Core-to-core network for the physics of self-organizing active matter, Jan. 29-30, Wenzhou, China [Invited talk]
- **Persistent random walk: a phenomenological paradigm for cell migration on solid substrates** Sep. 2023
The 7th International Soft Matter Conference, ISMC2023, Sep. 4-8, Osaka, Japan [Best Poster Award]
- **Curvature-promoted migration and alignment of cells inside microcylinders** Nov. 2022
CPS Fall Meeting-2022, Nov. 17-20, Shenzhen, China, Virtual
- **Persistent random walk: a phenomenological paradigm for cell migration on solid substrates** Oct. 2022
The 12th National Conference on Soft Matter and Biological Physics, Oct. 28-31, Wenzhou, China, Virtual
- **Variational methods and deep learning-based methods for active matter physics** Sep. 2022
EMBO Workshop: Physics of cells, Sep. 12-16, Ein Gedi, Israel, Virtual [Poster]
- **Anisotropic cell migration on stiff cylinders** Aug. 2021
The 6th National Conference of Statistical Physics and Complex Systems, Jul. 30-Aug. 2, Changchun, China, Virtual
- **Isotropic-to-Anisotropic transition of cell migration on cylinders induced by cell-scale curvatures** Jun. 2021
EMBO Workshop: Physics of living systems, Jun. 07-10, Virtual [Poster]
- **Curvotaxis of cells: Coarse-grained physical modeling** Aug. 2019
Chinese Congress of Theoretical And Applied Mechanics, Aug. 25-28, Hangzhou, China [Poster]

- **Phenomenological modeling of Curvotaxis of microdroplets and adherent cells** Jul. 2019
The 5th National Conference of Statistical Physics and Complex Syetems, Jul. 26-29, Anhui, China

OTHERS

- Languages Mandarin (Native), English (Fluent)
- Technical Mathematica, Python, Comsol, Fortran, Matlab, Gnuplot, LaTeX, Microsoft Office

REFERENCES

Prof. Xinpeng Xu

Associate Professor, Department of Physics, Guangdong Technion-Israel Institute of Technology, Guangdong, China

Email: xu.xinpeng@gtiit.edu.cn

Scholar Profiles: [Home Page of Xinpeng Xu](#)

Prof. Penger Tong

Chair Professor, Department of Physics, Hong Kong University of Science and Technology, Hong Kong

Email: penger@ust.hk

Scholar Profiles: [Home Page of Penger Tong](#)

Prof. Masao Doi

Chief Scientist, Wenzhou Institute, University of Chinese Academy of Science, China

Email: doi.masao.y3@a.mail.nagoya-u.ac.jp

Scholar Profiles: [Home Page of Masao Doi](#)

Prof. Tiezheng Qian

Professor, Department of Mathematics, The Hong Kong University of Science and Technology, Hongkong, China

Email: maqian@ust.hk

Scholar Profiles: [Home Page of Tiezheng Qian](#)

Prof. Dadong Yan

Professor, Department of Physics, Beijing Normal University, Beijing, China

Email: yandd@bnu.edu.cn

Scholar Profiles: [Home Page of Dadong Yan](#)